Executable Evidence Management for Aircraft Inspection

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Abstract

Inspection evidence can become inapplicable when a camera, detector, engineering requirement or output version changes. We present an executable interface that maintains quality statements through these updates. Typed requests compile to calibrated events; containment reuses compatible stronger events, exact fallback preserves direct-renewal support, and authenticated repair binds evidence to the current output. A CAD-association campaign uses 256 calibration scenes and 64 independent test scenes, each undergoing 16 operations including combined updates, stale delivery, repair and rollback. All complete methods support 54 of 64 fixed contexts. Containment reduces full-catalogue record evaluations from 13,824 to 8,960 (35.19%) and restores all 128 eligible repair checkpoints without an invalid binding. A second case executes four detector producers on public aircraft-defect imagery. Source-family and repeated-capture grouping reduces 10,722 released images to 1,339 representatives in 222 inferred source groups, with 33 calibration and 34 test groups. None of the 16 real-image requests passes the source-group calibration at allowance 0.25, so all quality statements are withheld while 68 authenticated report repairs preserve output integrity. The two cases separate detector or mapping quality, statistical applicability and report integrity within one reproducible evidence workflow.

Keywords: Engineering knowledge representation, Evidence reuse, Engineering change management, Risk calibration, Aircraft inspection

1. Introduction

An aircraft inspection image becomes useful when its observations can be located on the airframe and assessed against an engineering requirement. CAD-guided aircraft inspection establishes this spatial correspondence from model features [1]; fastener-inspection systems then evaluate surface components under the available imaging conditions [2]. During a repeated camera pass, increasing exposure can recover surface contrast while increasing motion blur. The airframe geometry remains unchanged, yet the processing configuration may associate a different set of fasteners with the CAD model. A quality statement established for the earlier configuration must therefore be checked before it accompanies the new output. The same issue arises when a defect detector is replaced: the new model can change recall and false positives even though the report format and aircraft task remain the same.

Such decisions accumulate within a continuing inspection workflow. A relaxed mapping tolerance may make an existing result usable, while a delayed output can arrive after its processing configuration has been replaced. Requirement-based digital-twin studies identify context, uncertainty, thresholds and historical information as inputs to maintenance decisions [3]. Selective sensor calibration concentrates effort on measurements identified as unreliable [4]. For inspection, the corresponding problem is to retain computations and quality statements that still apply, renew those affected by a change, and attach the result to the correct observation. The decision must survive successive updates and repairs, including a legitimate return to an earlier numerical configuration.

Existing research supplies substantial foundations for this process. CAD consistency analysis identifies repairs after model edits [5], lifecycle knowledge graphs preserve specification, version and operating-state histories [6], and dynamic safety cases update evidence when monitored conditions change [7]. Provenance

records describe how an output was generated and revised [8]. We study how these records support a quantitative inspection decision: whether a calibrated event covers the requested quality, whether the functions defining that event remain compatible, and whether the evidence is attached to the authenticated current output.

We develop an executable evidence interface that carries an inspection service through these changes. First, typed requests compile to quality events whose calibrated bounds can be reused by event containment; exact-event fallback preserves the services supported by direct renewal under the same calibration. Second, a versioned repair transaction restores the association to an authenticated output while retaining its change history and the event's admission decision. Third, a sequential change campaign evaluates these decisions against dependency-aware renewal and independent registry implementations, separating numerical computation from evidence preparation. A second case executes defect detection on public aircraft-skin images, using source groups to separate repeated captures from independent calibration trials. The two cases ask the same operational question: which requirement can the current producer and output support?

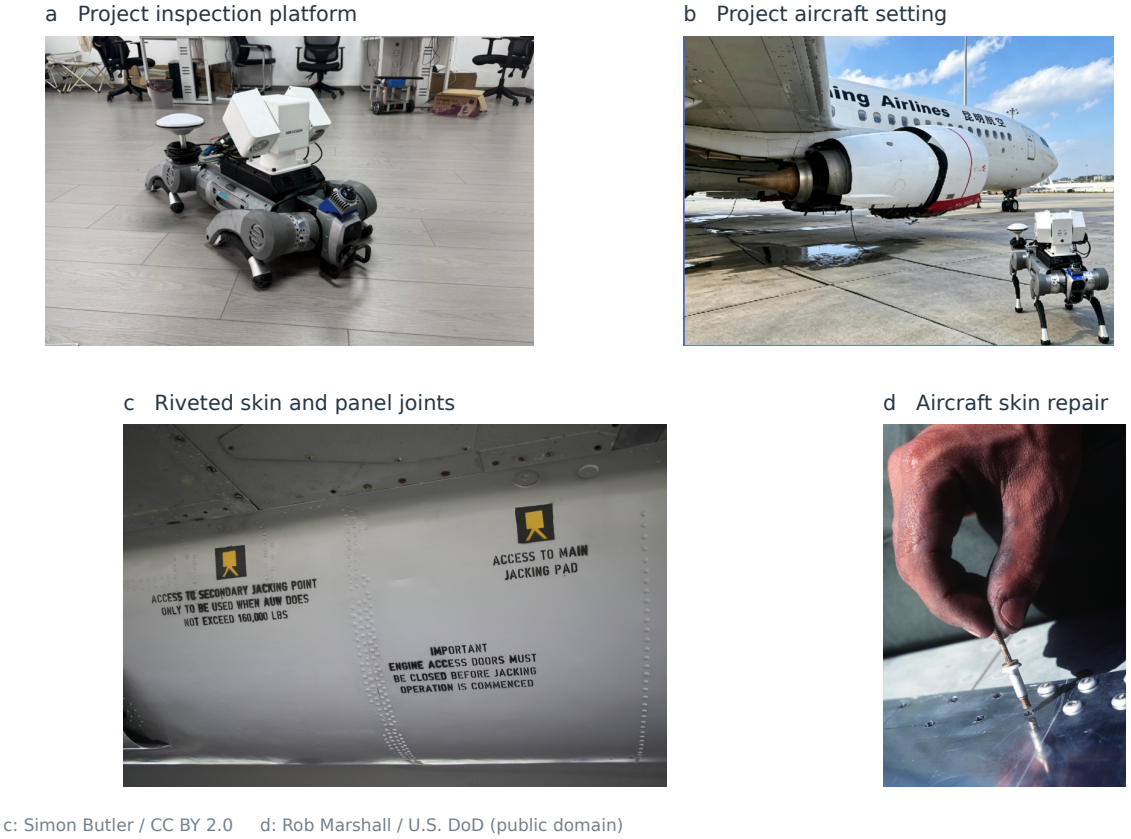


Figure 1: Aircraft inspection at platform and surface scales. (a,b) Original project photographs. (c) Under-fuselage panels on Avro Vulcan XM655, photographed by Simon Butler (CC BY 2.0). (d) Aircraft sheet-metal repair, photographed by Rob Marshall (U.S. Department of Defense, public domain). Sources and image credits are archived in the Supplementary Material.

2. Related work

2.1. Engineering knowledge and change-dependent decisions

Engineering knowledge representations connect requirements, model entities and decisions. Petnga and Austin integrate ontological reasoning with physical quantities and unit-aware computations [9]. Zou and

Feng resolve inconsistencies introduced by direct CAD editing through constraint analysis and ranked repair candidates [5]. These approaches establish the role of an actionable consistency decision. Quinn and McArthur assess representational completeness and expressiveness through industrial concept mappings and application relations [10]; inspection-robot ontologies relate task requirements to hardware capabilities [11].

Recent AEI systems connect engineering representations to executable models and lifecycle histories. Xiao et al. map design and building-model information into executable energy-analysis models [12]. Yang et al. separate persistent entities, specifications, released versions and runtime states, with explicit change actions and retained histories [6]. Our interface adopts this distinction between identity and changing state. Its maintained object is a quantitative quality event: the event's units, thresholds and producer dependencies determine which calibrated statement supports a request.

2.2. Provenance, selective recomputation and evolving evidence

The W3C PROV model describes generation, use, derivation, revision and invalidation; its constraints formalize consistency among these records [8, 13]. PDIFF compares execution provenance to locate differences caused by data, service or workflow revisions [14]. RECOMP studies selective recomputation as data and methods evolve [15], and version-aware restart trees identify affected outputs for recomputation [16]. Build-system theory supplies dependency-directed retention of unchanged computations [17]. These results motivate the dependency cache used by all matched incremental methods in this study.

Evidence maintenance also has an established assurance literature. Dynamic safety cases couple monitored variables, confidence structures and update rules, including removal of arguments whose assumptions become invalid [7]. SafeMachine organizes machine-safety assessment around traceable, context-aware evidence [18]. Assumption–guarantee contracts specify component obligations [19], and runtime-assurance architectures mediate permissible execution [20]. The present interface specializes evidence maintenance to a calibrated scene-quality event and the output receiving that statement. Its recovery transaction checks the current trusted binding; its event rule determines whether the requested service can resume.

2.3. Risk calibration and visual inspection

Learn then Test selects predictive programs using hypothesis tests with high-probability risk control [21]. Conformal Risk Control calibrates a monotone loss to control its expectation [22]. We use exact binomial inversion [23] with a fixed finite-family allocation. Event inclusion supplies an ordering for evidence reuse, and exact fallback reserves the original request's test. The resulting risk statement has a specified population, producer and quality event.

Industrial visual inspection also requires the event to reflect the task. Component-aware anomaly detection separates component structure and logical arrangement from local appearance [24]. CAD-guided aircraft inspection and rivet inspection provide the spatial and component-level setting for our capture-to-CAD requirements [1, 2]. Appearance-randomized rendering can improve transfer across image domains [25]; Xu et al. evaluate simulation-to-experiment adaptation for ultrasonic concrete-damage signals [26]. The controlled image study isolates correspondence and mapping quality, while public aircraft-skin defect imagery supplies a second task with annotated cracks, dents, missing fastener heads, paint loss and scratches [27]. We retain separate event definitions and calibration populations for these two services. A supplementary surface challenge evaluates light transport through explicit skin geometry, materials and illumination [28].

3. Inspection service and evidence model

3.1. A continuing inspection decision

Consider a camera observation used to associate fastener positions with CAD addresses. A complete-service request constrains spatial sampling, motion blur, correct-address coverage, fitted residual and physical mapping error. Increasing exposure preserves the geometric sampling scale but changes blur and image formation. A combined update to exposure, illumination and extraction changes several functions at once. The evidence interface determines which earlier statements remain applicable and constructs an exact-event statement when a compatible stronger one is unavailable.

Table 1: Closest research connections and the inspection decision examined here.

Research connection	Established capability	Inspection decision
Lifecycle knowledge [6]	Preserve identities, versions, states and change dependencies	Which requested quantities remain covered after a producer change?
Dynamic evidence [7]	Update arguments and evidence as monitored conditions change	Which quality event has an applicable numerical risk bound?
Workflow provenance [14, 16]	Locate changed computations and affected outputs	Which statements can accompany retained or corrected outputs?
Risk-controlled selection [21]	Admit programs under a calibrated risk allowance	How can reuse preserve exact-renewal support?

The output also carries an observation identity, producer signature and registry revision. An old output delivered after an update fails the current binding check. Recovery then verifies the trusted output and commits a newer revision. A subsequent rollback can restore an earlier numerical configuration while advancing the registry revision again. Figure 2 shows the relation between image quality and statistical admission for the first test source; the sequential experiment then follows its output through repair and rollback. The first held-out observation illustrates this distinction (Fig. 2). Its sampling scale remains 0.658 mm/pixel after the exposure update, while blur increases from 0.776 to 1.163 pixels. Both individual outputs satisfy the strict complete-service predicate. The independently calibrated upper bound nevertheless changes from 0.093 to 0.292, above the allowance 0.15. The relaxed complete event has upper bound 0.104 and remains supported. The workflow therefore continues the relaxed service while withholding the strict one; the output's current producer and binding determine which statement accompanies it.

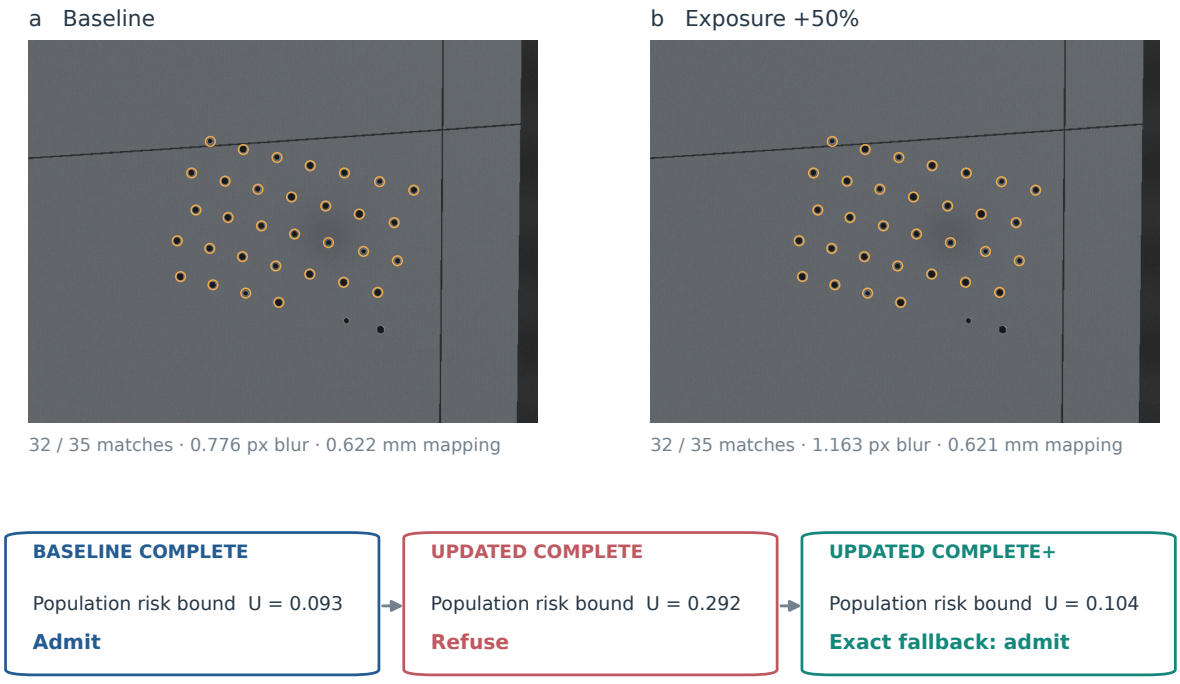


Figure 2: Actual controlled-raster inputs from the first held-out source before and after an exposure update. Orange rings mark centers extracted from pixels. Both particular images satisfy the strict quality event, while the independently calibrated population bounds support different service decisions. Exact fallback retains the relaxed complete service after the update.

3.2. Typed requests, events and dependencies

An engineering request is $R = (g, a, \varphi, \delta)$: population g, service a, quality predicate φ and allowable scene-level failure probability δ. Its grammar is

$$\varphi ::= \mathrm{Interval}(m, u, \ell, h) \mid \varphi \wedge \varphi, \tag{1}$$

where m identifies a quantity and u its engineering unit. Following unit-aware engineering knowledge models [9], compilation checks dimensions, converts units, intersects repeated bounds and canonicalizes the event. Nonfinite required values fail the quality predicate; an absent required field prevents evaluation as a schema error. The accepted set and its failure event are

$$S_R = \{z : \ell_{R,j} \le z_j \le h_{R,j},\ j \in J_R\}, \qquad E_{p,R} = \{z_p(X) \notin S_R\}, \tag{2}$$

where p identifies the numerical producer and X is a complete scene from population g.

Typed records connect the request to its evidence and output (Table 2). Their relation signatures include requires : Request $\to$ Event, bounds : Card $\to$ Event, dependsOn : Quantity $\to$ Provider and uses : Receipt $\to$ SourceVersion. In the schema, Receipt denotes the execution record. Schema validation checks relation endpoints; request compilation resolves the event's quantities and dependencies. The representation stores immutable specifications separately from active registry revisions.

For quantity j, the provider manifest declares dependencies D_j; the request uses $D_R = \bigcup_{j \in J_R} D_j$. Its producer projection identifies the corresponding registered functions and parameters. Within each source, a numerical stage also uses an ordinary content-addressed key [17],

$$h_v = \mathrm{Hash}\big(v, \sigma_v, \{h_u : u \in \mathrm{pa}(v)\}\big), \tag{3}$$

where σ_v identifies the stage implementation and settings. A cached value is retained when the registered stage and its inputs agree. The experiments fix implementation code within each run and maintain separate numerical caches for different sources. This stage cache controls numerical recomputation. The request projection separately selects the functions relevant to the quality statement. Before a component update is registered, a paired regression gate re-executes common inputs and rejects a declared invariance if an unaffected quantity changes. Runtime decisions use the registered signatures.

Table 2: Records connecting a quality requirement to its current output.

Record	Required content	Decision
Request	Population, service, units, intervals, allowance	Compile event and dependencies
Evidence card	Event, producer, calibration lineage, n, k, U	Cover the requested event
Source version	Observation identity, active producer, output digest	Authenticate the current binding
Repair transaction	Expected revision, trusted output and new revision	Restore a quarantined binding
Execution record	Plan, attachment and transaction history	Explain issuance or refusal

3.3. Image-to-CAD quality

Pixel extraction supplies candidate fastener centers. A pose prior projects nominal CAD locations under the perspective camera model [29], and minimum-cost assignment uses image-plane distances within a prior gate. The implementation uses the modified Jonker–Volgenant assignment routine described by Crouse [30]. For assigned correspondences $\mathcal{I}$, pose refinement solves

$$\begin{aligned}\widehat{\xi} = \arg\min_{\xi} \sum_{i \in \mathcal{I}} \sum_{d \in \{x,y\}} \rho\left(\frac{\pi_\xi(q_i)_d - u_{i,d}}{\sigma_u}\right) \\ + \sum_{d=1}^{3} \left[\rho\left(\frac{\Delta t_d}{\sigma_t}\right) + \rho\left(\frac{\Delta \theta_d}{\sigma_\theta}\right)\right].\end{aligned} \tag{4}$$

Here q_i is a nominal CAD address, u_i a pixel-derived center, and π_ξ the perspective projection. Translation and local rotation perturb the prior. The componentwise Huber loss [31] has transition 2. Estimation uses pixels, nominal CAD and the pose prior.

Offline scoring evaluates the correspondence and physical map after estimation. Correct-address coverage is the fraction of planned sites assigned to their physical counterparts. Fitted residual includes every assigned correspondence, including incorrect assignments. Maximum mapping error evaluates the estimated image-to-CAD map $H_{\widehat{\xi}}$ at every planned physical image location:

$$e_{\mathrm{CAD}} = \max_{i \in \mathcal{P}} \| H_{\widehat{\xi}}(u_i^\star) - q_i^\star \|. \tag{5}$$

Scene geometry supplies these evaluation labels after estimation. In the supplementary surface challenge, the map intersects the estimated camera ray with the exterior skin CAD mesh. Planned sites remain in the evaluation under occlusion, and a missing intersection gives infinite error. The observable image screen uses matched-site count and fitted residual; calibration and evaluation use every requested quality quantity, including correct-address coverage and mapping error.

3.4. Defect-report quality

The second service produces defect boxes and categories from an image. A producer identifies the detector weights, preprocessing, inference resolution and output threshold. A model update changes its weights; a digital gain update changes the input function. Both receive distinct producer signatures. The interface treats these outputs in the same way as a CAD-association packet, while using a different quality event.

For a source group G, let N_G be its image count, M_G its annotated defects, and $\mathrm{TP}_G, \mathrm{FP}_G$ its matched and unmatched predictions. Define

$$r_G = \frac{\mathrm{TP}_G}{M_G}, \qquad f_G = \frac{\mathrm{FP}_G}{N_G}, \qquad E_{p,R} = \{ r_G < r_R \ \vee \ f_G > f_R \}. \tag{6}$$

Recall r_G is dimensionless and f_G has units of detections per image. An empty-annotation group has recall one and is assessed through false positives. Class-aware one-to-one matching uses intersection-over-union at least 0.5; duplicate predictions are false positives. This event concerns pooled detection within a complete group. Its population, metric identity and thresholds distinguish it from a per-category detection guarantee. A request for a different metric therefore requires its own event and evidence.

4. Service maintenance through change

4.1. Validity of calibration-selected services

For k failures among n independent calibration units (complete synthetic scenes or complete real-image source groups), a one-sided Clopper–Pearson bound [23] is

$$U(k, n; \alpha) = \begin{cases} 1, & k = n, \\ \mathrm{Beta}^{-1}(1 - \alpha; k + 1, n - k), & k < n. \end{cases} \tag{7}$$

Fix a family of K producer/event functions before calibration and allocate one-sided error levels whose sum is at most β. The common confidence event then covers all K risks, using the familywise selection principle of Learn then Test [21]. The experiments use equal allocation $\alpha = \beta/K$. A service plan selected from these bounds is committed before evaluating a new scene.

A stronger requirement provides evidence for a weaker one: $S_A \subseteq S_R$ implies $E_{p,R} \subseteq E_{p,A}$. Producer transfer additionally requires identity of the requested functions,

$$z_{p',j}(x) = z_{p,j}(x) \quad \text{for all } j \in J_R \text{ and supported } x. \tag{8}$$

Registered dependency projections express this interface condition. The statistical guarantee is conditional on functional identity; the component regression gate tests the declared implementation on archived inputs.

Proposition 1 (Validity of calibration-selected services)**.** *Let $C \sim P^n$ be calibration data and let a new scene $X \sim P$ be independent of C. Fix K producer/event functions with one-sided allocations summing to at most β. With probability at least $1-\beta$ over C, every admitted request selected using C has fresh-scene risk at most δ, provided that its event is pointwise contained in an anchor event from this fixed family with $U_A(C) \le \delta$ and its transferred functions satisfy Eq. (8).*

Proof. The union bound gives a calibration event of probability at least $1-\beta$ on which every fixed-family risk is bounded by its $U_A(C)$. On that event, pointwise inclusion gives $\Pr_X(E_R \mid C) \le \Pr_X(E_A \mid C) \le U_A(C) \le \delta$ for every eligible selection. Requested-function identity preserves the inclusion after transfer. □

The proposition concerns the quality of a planned service on a fresh scene. For a subsequent issuance indicator I, it also gives $\Pr(E_R \cap \{I=1\} \mid C) \le \delta$ on the same calibration event. The conditional failure rate among outputs accepted by a scene-dependent screen, $\Pr(E_R \mid I=1, C)$, is a separate quantity. The supplementary surface challenge separately measures screen acceptance and accepted-but-failed outcomes. The defect service uses the complete-group event in Eq. (6), without selecting test groups by their observed detection quality.

4.2. Containment with exact-event fallback

For a committed producer, the interface first searches compatible covering cards with $U_A \le \delta$. If none qualifies, it constructs the request's exact-event card from the fixed family. The route is

$$Q(R) = \begin{cases} \text{reuse}(A), & \mathcal{A}(R) \neq \varnothing, \\ \text{exact}(R), & \mathcal{A}(R) = \varnothing,\ U_R \le \delta, \\ \text{refuse}, & \text{otherwise.} \end{cases} \tag{9}$$

Here $\mathcal{A}(R)$ contains cards with matching population and service, compatible producer projection, and a stronger accepted set whose bound satisfies $U_A \le \delta$. Unsuccessful anchors and refusal cards count toward construction work.

Proposition 2 (Support preservation with exact fallback)**.** *Suppose the compared cards use the same n calibration scenes and allocation α, and transferred request quantities agree on those scenes. Containment followed by exact-event fallback admits exactly the requests admitted by the complete exact-event catalogue.*

Proof. Containment gives $k_R \le k_A$. At fixed n and α, the binomial upper bound is nondecreasing in k, so admission by an anchor implies $U_R \le U_A \le \delta$. Otherwise, fallback evaluates the same exact event as the complete catalogue. □

The fixed family reserves an allocation for every exact fallback. A failed strict anchor can therefore lead to an admissible relaxed event within the original confidence budget. A change in registered population identity selects a different evidence population before any transfer decision.

4.3. Quarantine, repair and revision order

Provenance distinguishes the generation, revision and invalidation of an output [8, 13]. The output binding identifies a source, producer, version and payload digest. An authoritative registry stores the current tuple. Issuance requires both an admissible plan and agreement with that tuple. An invalid attachment places its binding workflow in quarantine and records the rejected attempt. Ordinary retries preserve the quarantine.

Recovery uses an explicit transaction with the expected registry revision, the trusted active output and the next registry revision. The registry verifies the producer, payload digest, source lineage and calibration identity before accepting the transaction. A successful repair replaces the binding record and retains its history. A missing repair certificate, wrong digest, replayed transaction or unregistered alias is rejected. An invalid command preserves any previously valid committed state.

The resulting invariant is that every issued binding equals the current authenticated registry record, and every transition out of quarantine has a successful repair transaction in its history. Revision order belongs to this control history. Restoring an earlier numerical configuration therefore creates a new revision; it does not authorize a late attachment carrying the old revision. The event card continues to determine whether that producer supports the current request after repair or rollback.

5. Experimental design

5.1. Scene population and calibrated requirements

The lifecycle experiment samples 256 independent calibration scenes and 64 independent test scenes from a controlled image population. Each scene draws its own surface curvature, camera pose, material appearance and physical-to-CAD displacements. Four fastener layouts are represented: parallel rows, curved rows, staggered sites and an access-panel perimeter. The layouts contain 28–36 sites, with fastener radii of 2.0–3.4 mm and typical spacing of 28–40 mm. Shared and local displacement standard deviations are 0.4 and 0.15 mm; site missingness is 0.025 and edge occlusion occupies 0–10% of the image.

Baseline images have 768×576 pixels, focal length 1,200–1,600 pixels, camera distance 0.70–1.15 m, exposure 3–7 ms and surface-relative motion 0.06–0.18 m/s. The family contains the baseline and five component updates: focal length $\times 1.15$, exposure $\times 1.5$, image region 896×672, illumination $\times 0.45$ and extraction contrast threshold $18 \to 10$. Two combined producers apply exposure/illumination/extraction together and focal length/image region together. Named configurations are complete settings; only the two explicitly combined producers apply simultaneous changes.

All eight producers execute on every calibration scene, yielding 2,048 outputs. The eight requests in Table 3 give the fixed family $K = 8 \times 8 = 64$, with $\beta = 0.05$, $\alpha = 0.05/64$ and primary allowance $\delta = 0.15$. The allowance concerns violation of the capture-to-CAD quality conjunction by a complete scene. Calibration selects the plans before the test inputs are processed. The six producers visited by the sequential campaign execute on all 64 test scenes; evaluation scores all eight requests for each of these producers, giving 48 observed producer/request contexts.

Table 3: Capture-to-CAD requirements. GSD is in mm/pixel; blur and residual are in pixels; coverage is in percent; mapping error is in mm. A dash leaves the quantity unconstrained; + identifies a relaxed request.

Request	GSD $\leq$	Blur $\leq$	Coverage $\geq$	Residual $\leq$	Mapping $\leq$
Optical	1.2	3	—	—	—
Optical+	1.5	4	—	—	—
Acquisition	1	2	80	—	—
Acquisition+	1.2	3	60	—	—
Association	—	—	80	8	8
Association+	—	—	60	12	12
Complete	1	2	80	8	8
Complete+	1.2	3	60	12	12

Local dark contrast and connected components of 5–240 pixels supply candidate centers. A 30-pixel prior gate constrains CAD assignment; six or more matches trigger pose refinement with scales of 0.5 pixel, 3 mm and 0.001 rad. The implementation uses SciPy's trust-region reflective least-squares solver [32] with the specified Huber loss. The numerical dependency graph separates geometry, radiance, sensor formation, extraction, pose and optical quantities.

5.2. One continuing change campaign and matched methods

Each test source follows 16 operations: initial service, request relaxation, exposure update, request tightening, illumination update, extraction update, combined exposure/illumination/extraction update, combined camera update, stale attachment, out-of-order repair, valid repair, repair replay, numerical rollback, population mismatch, late old-version delivery and valid rollback repair. The illumination step requests the

optical event; other ordinary steps request strict or relaxed complete service. The population-mismatch step executes a paired stress input with 0.14 missingness, 0–35% occlusion and larger CAD displacements. Its registered population differs from the calibrated population.

Whole renewal recomputes the complete numerical packet and constructs exact-event evidence. The primary baseline uses a persistent dependency graph, exact-event cards and the authoritative registry. It reuses an exact card only when the event and its producer projection match; containment additionally permits a stronger event to cover a weaker request. Containment with repair uses the same graph and registry with stronger-event reuse and exact fallback. Independent rules implement the same event and binding conditions. All methods receive identical inputs, trusted outputs and calibration records. Stale delivery and repair operate on the actual stored numerical packets, including their image and correspondence hashes. The independent checker reconstructs the required revision, binding and admission for every operation.

A complementary component-registration experiment uses 64 additional independent scenes. It compares clean updates with three conditional hidden couplings: image region to focal length, illumination to exposure, and extraction to focal length. Every pair re-executes the affected numerical producer; changes beyond 10^{-10} absolute/relative tolerance veto a declared invariance. This experiment tests the dependency interface before the sequential campaign applies the registered configurations.

Evidence preparation has two demand schedules: all 64 fixed contexts and the requests visited by the 16-step campaign. Each schedule builds one population-level catalogue per method, shared by all test sources. We measure its constructed cards, calibration-record evaluations and process CPU. Numerical workflow timing separately measures the 64 paired trajectories, with method order rotated across sources. The complete packet is requested at each numerical step; stage caching and evidence reuse therefore have distinct work counters. All unsuccessful cards and refused requests remain in the measurements.

The experiment runs on a shared Linux AMD EPYC 9V74 host, Python 3.12, with two workers and one numerical thread per worker. Process CPU and elapsed time include executed numerical stages, geometric quality scoring, payload hashing and registry/consistency checks. Initial nominal-scene sampling, process creation and file export are excluded; the stress-input draw is included in its population-change step. Catalogue preparation is timed separately. A 10,000-replicate paired bootstrap resamples the 64 complete sources for the workflow CPU comparison. The Supplementary Material records the environment, all event outcomes and the earlier single-update and repair studies.

5.3. Real-image defect detection and report updates

The second case uses Innovation Hangar v2, version 1, distributed with the public aircraft-defect study of Suvittawat et al. [27]. Its categories are crack, dent, missing fastener head, paint loss and scratch. The released 10,722 images contain augmentation copies and repeated captures. We first retain one representative for each source-photo family, group dated captures by complete day, and merge additional near-duplicates by image correlation or geometric feature agreement. The resulting 1,339 representative images belong to 222 inferred source groups. Whole groups are assigned to training, development, calibration and test; the corresponding image counts are 458, 34, 368 and 479, and group counts are 122, 33, 33 and 34. Aircraft identities are unavailable, so these are inferred source groups rather than identified airframes. Table 4 records the retained evidence units.

A COCO-pretrained YOLO11n detector [33] is trained on the training partition. The early producer uses epoch 4. Further training and the final checkpoint are selected using development data alone, where crack and dent are the annotated categories; the final prediction threshold is 0.10. Inference resizes the 640-pixel released inputs to 480 pixels. The other two producers apply digital gain 0.5 or 1.5 to the same test pixels before running the updated model. These are controlled digital processing changes. The existing typed compiler registers group recall and false positives per image as two additional quantities; the common containment and authorization API checks both cases, including the archived CAD cards. Detection receives pixels only; labels enter subsequent scoring. We report bounding-box AP50 and AP50–95 with the COCO evaluation implementation [34], together with counts at the fixed operational threshold.

Four nested requests set (r_R, f_R) to $(0.25, 2)$, $(0.50, 1)$, $(0.75, 0.50)$ and $(0.90, 0.25)$, named Triage, Review, Coverage and Strict. Each source group contributes one Bernoulli outcome to Eq. (6); repeated images are

pooled within that group. The fixed family has four producers and four events, $K = 16$, with $\beta = 0.05$ and primary group-failure allowance $\delta = 0.25$. The confidence calculation assumes independent source-group draws from the stated population. Its application to the inferred groups is assessed through the recorded grouping audit. We also report allowances 0.10 and 0.40, and 100 subsets of 24 calibration groups as a descriptive sensitivity analysis. A threshold query $(0.60, 0.75)$ outside the fixed family uses a covering allocated event or is refused.

Each of 34 test groups then follows 12 report operations: early-model output, model update, request tightening, stale delivery, repair, darkened producer, request relaxation, brightened producer, numerical rollback, late old-version output, repair and repair replay. Payload digests include the actual image and detection hashes. The same authoritative registry and independently folded history used in the CAD case check every operation. Training and inference execute on a four-core Linux CPU allocation; their measured costs are reported separately from the CAD campaign timings and from evidence-card work. The Supplementary Material supplies per-class results, the complete split manifest, training selection and all report transitions.

Table 4: Retained real-image partitions. Source groups, rather than augmented images or boxes, define the calibration unit.

Partition	Source groups	Representative images	Defect instances
Training	122	458	870
Development	33	34	107
Calibration	33	368	614
Test	34	479	641

6. Results

6.1. The supported service changes with the requirement

Calibration supports 54 of the 64 producer/request contexts (Fig. 3). All four complete methods retain this same set. Under the fixed catalogue order, containment supplies 29 contexts from existing stronger cards; exact construction supports 25 further contexts and refuses ten. The exposure update raises the strict complete-event bound to 0.292, while its relaxed event remains at 0.104. The combined exposure, illumination and extraction update similarly supports relaxed complete service at 0.130 while refusing strict complete service at 0.352. Under reduced illumination, the optical event remains supported at 0.028; complete association requirements determine the additional refusals.

The six producers used in the sequential test yield 48 observed contexts, of which 38 were admitted by calibration. Among these 38 calibration-admitted contexts, test failures range from 0 to 4 per 64 scenes. At baseline, mean correct-address coverage is 95.94%, median fitted residual is 0.646 pixel, and median mapping error is 0.586 mm. Median prior translation error decreases from 4.431 to 1.254 mm after fitting. All test outcomes, including the refused programs and individual quality failures, are retained in the Supplementary Material.

In the separate 64-source component-registration experiment (Supplementary Material), paired numerical re-execution detects all 86 activated hidden couplings, with no veto among its 192 clean update pairs. These checks exercise numerical effects of a violated dependency declaration. The sequential experiment then examines the evidence and output history of the registered updates.

6.2. Continuing service through repair and rollback

The campaign distinguishes three causes of refusal within the same output history (Fig. 4). Tightening the exposure-updated request fails its risk allowance despite a current authenticated binding. Stale delivery and an out-of-order repair encounter an admissible event but fail the binding rules. The stress input has a different registered population, so its quality statement is unavailable before any attachment decision. These decisions follow from the event, producer and output records supplied to each method.

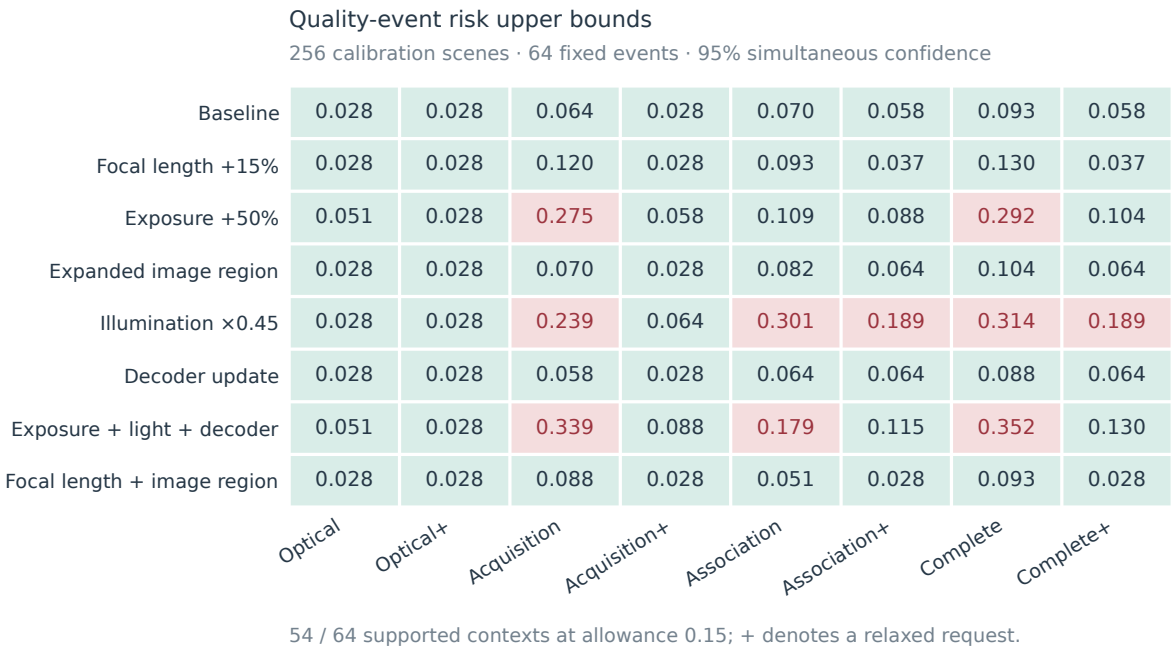


Figure 3: Exact-event calibration bounds for the 64 fixed contexts. Green cells meet the 0.15 scene-level allowance. The 256 calibration scenes are independent of the 64 campaign test sources. The three incremental methods and whole renewal support the same 54 contexts.

All four methods make the same 704 issuance decisions among 1,024 operations per implementation. No stale, wrong-origin or invalid-revision binding is issued. Both authorized repairs succeed for each of the 64 sources, restoring all 128 eligible recovery checkpoints. A replayed repair is rejected while the already repaired output remains available. Numerical rollback restores the cached baseline output at registry revision 9, advancing from revision 8; the incremental methods execute no numerical provider for that operation. A late delivery carrying the earlier revision is still refused, and the final repair restores service for all 64 sources. Thus reuse of numerical data and reuse of control authority follow different revision rules.

6.3. Evidence work and numerical computation

For the complete family, whole renewal constructs 64 cards, exact dependency caching constructs 54, and containment with fallback constructs 35. The corresponding record evaluations are 16,384, 13,824 and 8,960. Containment reduces this work by 35.19% relative to exact caching while preserving all 54 supported contexts. The independent rules give the same construction and support counts. For the requests actually visited in the 16-step campaign, exact caching constructs eight cards and containment constructs six: 2,048 versus 1,536 record evaluations, a 25% reduction. Each catalogue is prepared once and shared by all 64 sources.

Figure 5 separates this evidence work from numerical execution. Whole renewal consumes 193.617 s of process CPU across the 64 trajectories, compared with 78.773 s for the DAG baseline. The shared dependency cache reduces workflow CPU by 59.31%, with a paired 95% bootstrap interval of 58.50–60.12%. Containment and rules consume 79.328 and 79.687 s, with identical numerical outputs and stage counts. A separate cold full-catalogue preparation takes 9.577 ms for exact caching and 6.324 ms for containment. The principal operational result is selective continuation and recovery with explicit evidence applicability; the additional containment saving concerns this small preparation task.

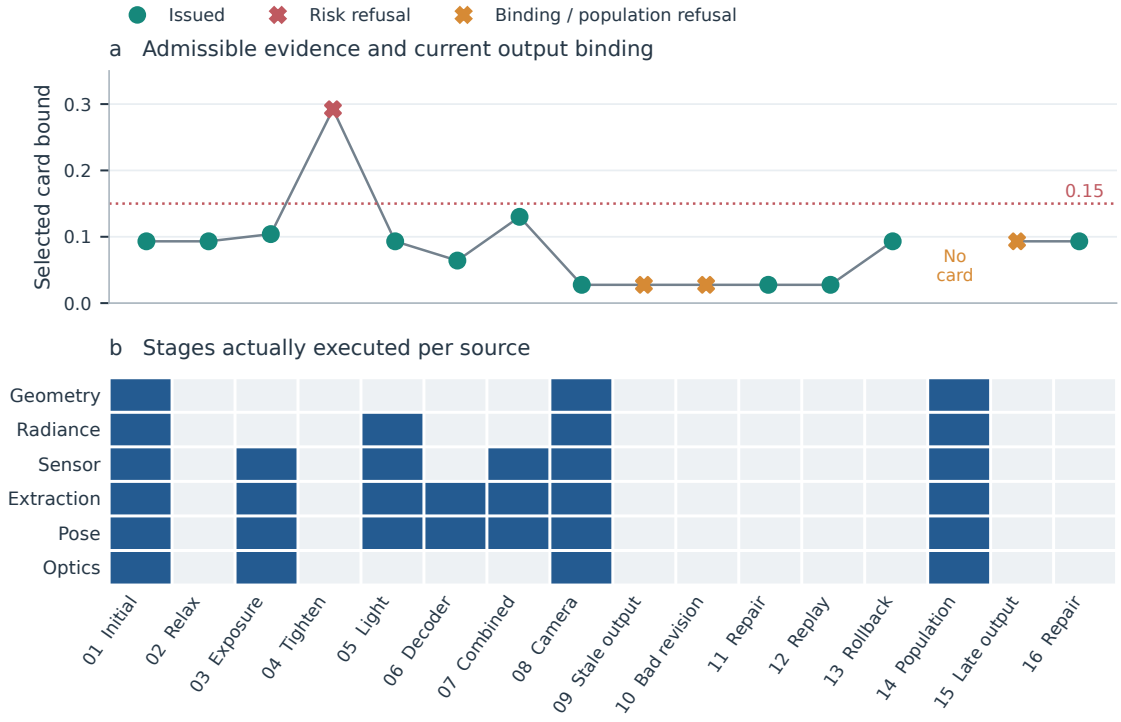


Figure 4: The 16-operation lifecycle. (a) The selected evidence bound and actual issuance decision. Reused stronger cards can have a higher bound than the request's exact card. A statistically supported service can still be withheld for a stale binding; a population mismatch has no applicable card. (b) Actual dependency-stage execution. Valid repair and numerical rollback use retained outputs. The independent registry and rule implementations reproduce these decisions and stage counts.

6.4. Real-image quality, admission and report integrity

The updated detector reaches AP50 17.1% and AP50–95 7.5% on 479 test images. At confidence 0.10 it recovers 157 of 641 annotated defects, with 726 false positives. Table 5 separates all four numerical producers. Figure 7 shows fixed-identifier examples containing each defect category, with both annotations and actual predictions. These image-level scores describe the detector; source-group event failures determine service admission.

Across the 16 fixed producer/request contexts, calibration observes 28–33 failures among 33 groups. The resulting simultaneous upper bounds admit 0 contexts at allowance 0.25 (Fig. 6). Exact renewal and containment with fallback agree for every context; both retain every failed group. The threshold query outside the reserved family is handled by containment when possible and refused otherwise. Every observed group-failure frequency already exceeds the allowance before confidence adjustment. The measured detection quality therefore drives refusal. With no qualified stronger anchor available, both exact renewal and containment evaluate the same 528 calibration-group records. The three less represented categories each appear in only one calibration and one test source group. Their frame counts do not provide a class-specific population guarantee: even zero failures in one independent group has unadjusted 95% upper bound 0.95.

The report campaign executes 408 operations on the 34 test groups. All 68 authenticated repairs commit a new registry revision, while replayed transactions and all stale-output attempts remain rejected. No quality-qualified report is issued. The repairs restore binding integrity while the current statistical admission remains unavailable. These report transitions are protocol checks; the CAD campaign supplies the positive service-recovery cases.

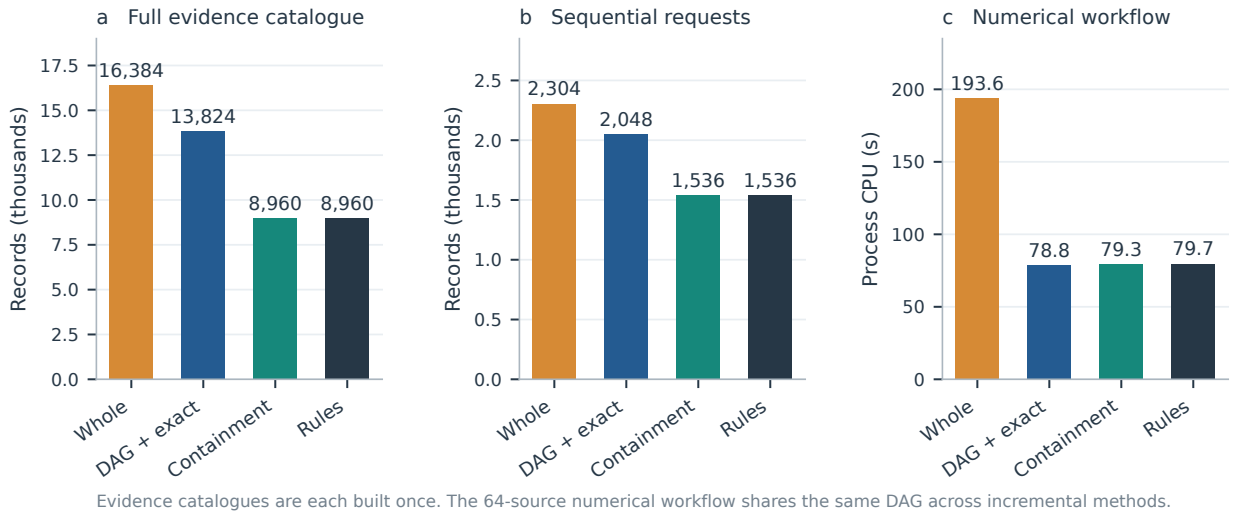


Figure 5: Attribution of computational work. The complete family and the sequential demand each build one shared evidence catalogue. Event containment reduces record evaluations relative to exact caching. The larger workflow CPU saving comes from the ordinary dependency graph shared by the three incremental methods.

Table 5: Actual detector results on the shared real-image test partition. Recall and false positives use the development-selected fixed confidence; AP uses the archived candidate detections.

Producer	AP50 (%)	AP50–95 (%)	TP	FP	Recall (%)
Early model	2.8	0.9	64	2675	10.0
Updated model	17.1	7.5	157	726	24.5
Gain 0.5	19.2	8.2	155	635	24.2
Gain 1.5	12.5	6.0	132	904	20.6

7. Discussion

The proposed interface maintains the applicability of a quantitative inspection statement through engineering change. Lifecycle knowledge graphs preserve the relationships among identities, specifications and operating states [6], and dynamic safety cases update evidence as monitored conditions change [7]. Here the maintained object is a calibrated quality event attached to a particular output. Requirement containment, producer compatibility and the current registry binding jointly determine whether the service is retained, renewed, refused or recovered. The sequential campaign places these decisions within one output history, including combined producer updates, repair replay and a numerical rollback at a newer registry revision.

The matched comparisons identify the role of each mechanism. Dependency caching retains unaffected numerical stages, following selective recomputation [15, 17]. Event containment reduces the records needed to prepare a quality statement, while exact fallback preserves the admission decision of the complete catalogue. Registry transactions determine which numerical output can carry that statement. The separate work measurements make these effects assessable under the same producer, calibration inputs and authoritative records. Independent rule and registry implementations test whether the decisions follow from the specified semantics.

Physical quality and output integrity have complementary roles. Correcting a stale attachment restores the association to a trusted output; the event's calibrated bound determines whether the requested service can resume. An output with acceptable fitted residual can still contain incorrect CAD associations. Likewise, good average detector precision does not establish a bound on the fraction of source groups that violate a

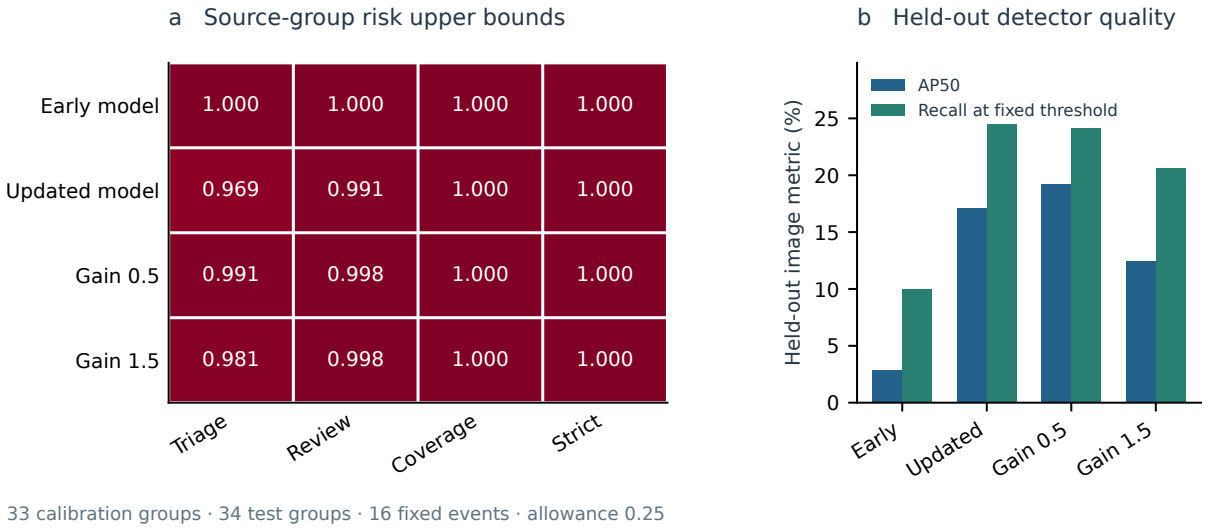


Figure 6: Real-image evidence at two units of analysis. (a) Each bound uses 33 complete calibration source groups with allocation 0.05/16. (b) AP and recall summarize detections on the shared 479-image test partition. The image summaries and source-group risk bounds describe different quantities.

requested recall and false-positive conjunction. The two cases make the quality quantity and its statistical unit explicit before comparing producer versions. The supplementary surface challenge further examines appearance, view and occlusion with the same planned CAD sites retained in the scoring denominator.

The defect case also makes the cost of a realistic data unit visible. Augmented images and neighboring frames increase the amount of image processing without providing an equal number of new calibration trials. Source-group evaluation defines the admission unit. In the executed real-image case, high observed failure frequencies drive refusal even before confidence adjustment; limited source diversity additionally restricts class-specific statements. Class-specific requests remain distinct from a pooled group request.

The workflow operates with declared provider dependencies and a trusted registry. Functional identity supplies the transfer condition, and paired regression checks its implementation on executed inputs. Population identity keeps each calibrated statement attached to its operating distribution. These explicit conditions make a renewed or repaired output inspectable: its record identifies the requested quality, the producer establishing that quality and the current output carrying the evidence.

8. Conclusion

This study connects engineering requirements, calibrated quality events and current inspection outputs through an executable evidence interface. Event containment reduces renewal work, exact fallback preserves the support of the fixed catalogue, and authenticated repair restores the output binding without reversing registry history.

The CAD campaign retains all 54 supported contexts while reducing full-catalogue evidence evaluations by 35.19% and sequential-demand evaluations by 25%. Each complete implementation restores its 128 eligible repair checkpoints with no invalid binding. The real-image case applies the same interface to actual defect detections, treating inferred source groups as the evidence unit and keeping detector performance separate from admission and report integrity. The measured quality fails all 16 requests, while authenticated repairs restore report integrity. The two outcomes remain separate throughout the workflow. The resulting workflow makes retention, renewal, refusal and recovery explicit decisions with traceable numerical inputs.

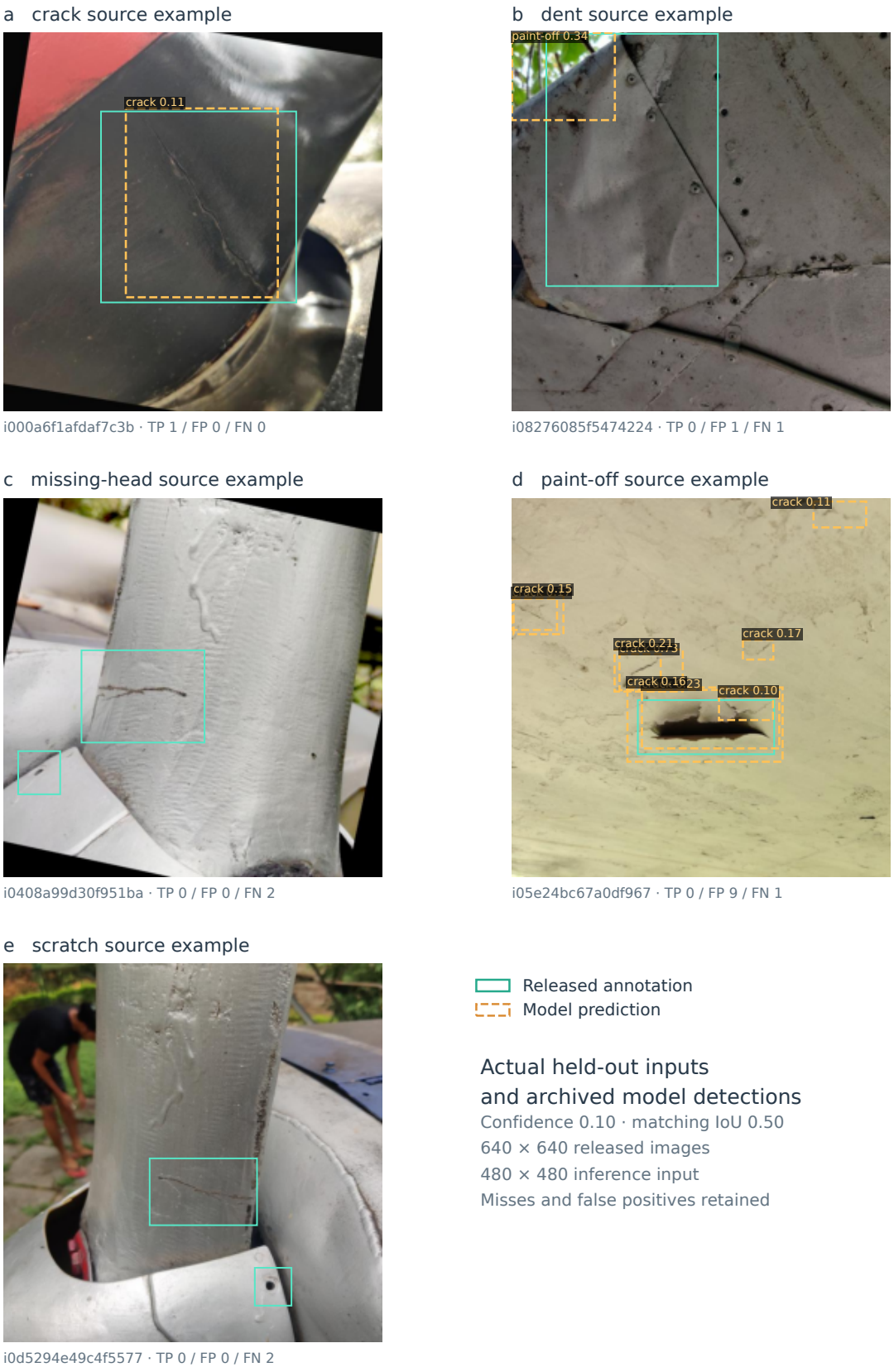


Figure 7: Actual held-out defect images and updated-model outputs. Solid green boxes are released annotations; dashed orange boxes are model predictions with category and confidence. Each panel uses the first unused test identifier containing its named category. TP, FP and FN count matched predictions, unmatched predictions and missed annotations for the complete panel.

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